

Dec 8, 2025 (Due: 08:00 Dec 15, 2025)

1. Let $A \in \mathbb{C}^{n \times n}$ be Hermitian, and $Q \in \mathbb{C}^{n \times k}$ has orthonormal columns (i.e., $Q^*Q = I_k$). Show that

$$Q^*AQ = \arg \min_{D \in \mathbb{C}^{k \times k}} \|AQ - QD\|_F.$$

(optional) Show that

$$Q^*AQ = \arg \min_{D \in \mathbb{C}^{k \times k}} \|AQ - QD\|_2.$$

Remark: This exercise can be interpreted as the optimality of the Rayleigh–Ritz procedure.

2. Implement the Lanczos algorithm for symmetric eigenvalue problems. Use MGS2 (MGS with reorthogonalization) to implement the Lanczos process (in fact, the Arnoldi process). Test the eigensolver on a relatively large-scale diagonal matrix with a dense starting vector. Visualize the convergence history for the algebraically largest eigenvalue and the corresponding eigenvector. Does your computational result agree with the theoretical prediction that $|\hat{\lambda} - \lambda| = O(\|\hat{x} - x\|_2^2)$?

Note: Since the true eigenpairs are known, it is possible to calculate the forward error for the computed eigenpairs.

3. Implement the Lanczos algorithm for symmetric eigenvalue problems, now using the standard three-term recurrence. Test is on a relatively large-scale diagonal matrix with a dense starting vector.

What do you observe for the convergence of Ritz pairs and the orthogonality of the Lanczos vectors?

4. Let A be a Hermitian positive definite matrix. Show that the iterative scheme

$$r_k = Ax_k - x_k \frac{x_k^* Ax_k}{x_k^* x_k}, \quad x_{k+1} = x_k - A^{-1} r_k, \quad (k = 0, 1, 2, \dots)$$

can be used to compute the smallest eigenvalue of A .

(optional) Try the preconditioned version

$$x_{k+1} = x_k - M^{-1} r_k,$$

for $M \approx A$ on some examples. What do you observe?

5. (optional) Implement the Krylov–Schur algorithm and use it to compute a few dominant eigenvalue of a nonsymmetric matrix.

6. (H) Write a program to compute the spectral decomposition of Wilkinson matrices. Visualize the decay behavior of the eigenvectors.